

SATHWIK RUPIREDDY

Data Engineer II

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PROFESSIONAL SUMMARY

Data Engineer with 4+ years of experience designing, developing, and maintaining scalable ETL/ELT data pipelines using Python, SQL, and Apache Airflow, with strong experience building and optimizing data warehouses and data lakes on AWS (Redshift, S3, Glue). Proficient in optimizing SQL queries, Redshift schemas, and data models for performance and scalability, with hands-on experience preparing datasets for machine learning forecasting models and implementing CI/CD pipelines for data workflows. Skilled in monitoring pipeline performance, troubleshooting production issues, maintaining data quality and governance standards, and participating in code reviews to improve engineering standards across cross-functional data platform teams.

TECHNICAL SKILLS

- **Python & SQL:** Python (strong, 4+ years), SQL (strong proficiency), data processing and pipeline development, query optimization
- **AWS Data Analytics Services:** AWS Redshift, AWS S3, AWS Glue, AWS EMR, AWS Lambda, AWS Step Functions, AWS MWA, AWS IAM
- **ETL/ELT Pipelines:** Scalable ETL/ELT data pipeline design and development, Apache Airflow (workflow orchestration), Python-based pipeline automation
- **Data Warehousing & Modeling:** Data warehouses and data lakes, Redshift schema optimization, data modeling, dimensional modeling, performance optimization, Snowflake
- **ML Data Pipelines:** Machine learning data pipelines, feature engineering workflows, forecasting use case dataset preparation, MLflow
- **CI/CD & Data Governance:** CI/CD pipelines for data workflows, infrastructure deployments, Git, Terraform, data governance and security best practices, documentation
- **Cloud Platforms:** AWS, Azure (Data Lake, Databricks), GCP, Docker, Kubernetes
- **Additional:** Apache Spark, PySpark, Apache Kafka, dbt, Power BI, Tableau, Great Expectations

PROFESSIONAL EXPERIENCE

CAI International | Data Engineer

Jul 2025 – Present | United States

- **Design, develop, and maintain scalable ETL/ELT data pipelines using Python, SQL, and Apache Airflow** — processing 50M+ records daily while improving analytics readiness by 42% across 12 production data initiatives.
- **Build and optimize data warehouses and data lakes using AWS Redshift, S3, and AWS Glue** — developing integration workflows and managing data infrastructure that supports analytics, reporting, and machine learning initiatives.
- **Optimize SQL queries, Redshift schemas, and data models for performance and scalability** — reducing data latency by 45% and lowering annual cloud expenditure by \$320K through systematic performance tuning.
- **Prepare and deliver datasets for machine learning forecasting models** — partnering with data science teams on feature engineering workflows and ML-ready data pipeline preparation.
- **Implement CI/CD pipelines for data workflows and infrastructure deployments** — reducing deployment timelines by 80% and increasing release frequency 4x via automated pipelines.
- **Monitor pipeline performance and troubleshoot production issues** — leading root cause analysis that reduced data-quality incidents by 35% and maintained 99.9% platform reliability.
- **Maintain documentation and adhere to data governance and security best practices** — participating in code reviews and contributing to engineering standards across a cross-functional team of 6.

Wipro | Data Engineer

Jan 2021 – Nov 2023 | India

- **Designed and maintained scalable ETL/ELT pipelines using Python and Apache Airflow** — processing 4.8TB+ of daily transactional data and improving pipeline efficiency by 41% across multiple business domains.
- **Built and optimized data warehouses and data models** — optimizing SQL queries and dimensional schemas to produce analytics-ready structures for BI and reporting consumers.

- **Monitored pipeline performance and troubleshoot production issues** through root cause analysis — reducing execution windows by 38% and improving SLA compliance to 99.5%.
- **Developed and managed data integration workflows using Apache Airflow** with 150+ automated production pipelines — improving stability to 99.7% and reducing manual effort by 70%.
- **Implemented CI/CD pipelines and maintained data governance documentation** — contributing to code reviews and engineering standards across distributed teams.
- **Built Python data clients and REST API integrations** handling JSON and XML formats across cloud-based distributed systems, sustaining 99.7% pipeline stability.

PROJECTS

- **Retail Sales Analytics Platform — AWS-Style ETL & Redshift-Optimized Data Warehouse** | dbt, Snowflake, Power BI — Designed scalable ETL pipelines and optimized data warehouse schemas in Snowflake/Redshift-equivalent architecture; built ML-ready datasets and Power BI dashboards reducing manual reporting effort by 60%.
- **Laptop Price Intelligence Pipeline — Cloud Data Lake & Forecasting Dataset Preparation** | PostgreSQL, BigQuery, Power BI — Engineered a cloud-based ETL pipeline on AWS-style architecture, preparing datasets for forecasting use cases and building interactive dashboards analyzing trends across thousands of products.

EDUCATION

Master of Science in Data Analytics | Webster University | GPA: 3.56/4.0 Dec 2025
Bachelor of Technology, Electrical & Electronics Engineering | Kakatiya University

CERTIFICATIONS

Microsoft Azure Database Administrator Associate | Google Data Analytics Professional Certificate | Microsoft Power BI Data Analytics & DAX | SQL for Data Analysis | Anthropic AI Fluency Certification | Claude 101 (Anthropic)